DATASET INTELLIGENCE & OPERATIONAL ANALYSIS

Domain: Universal Dataset Intelligence & Distribution Analysis (0.0%) | Profiling Conf: 90.0% | Records: 400 | Hash: 4f480840

Filter: ALL_RECORDS(UNFILTERED) | Audit: VERIFIED_DETERMINISTIC_PARITY | Engine: v8-javascript

KEY PERFORMANCE INDICATORS (DETERMINISTIC METRIC AUDIT)

TOTAL RECORDS VOLUME

400

`COUNTROWS`

Omitted: 0 rows

UNIQUE ENTITIES (SKU_CODE)

400

`COUNTDISTINCT(SKU_Code)`

Omitted: 0 rows

MEAN UNIT_COST

54.73

`AVERAGE(Unit_Cost)`

Omitted: 0 rows

MEDIAN UNIT_COST

54.50

`MEDIAN(Unit_Cost)`

Omitted: 0 rows

MIN UNIT_COST

12.50

`MIN(Unit_Cost)`

Omitted: 0 rows

MAX UNIT_COST

96.50

`MAX(Unit_Cost)`

Omitted: 0 rows

EXECUTIVE MANAGEMENT DECISION BRIEF

- **Ingestion & Data Quality Profile:** Audited 400 source rows across 8 columns with 100.0% full record completeness. Detected 0 duplicate records in source data.
- **Total Records Volume Benchmark:** Total Records Volume stands at 400 calculated via COUNTROWS. Supporting benchmark Mean Unit_Cost stands at 54.73.
- **Cohort Concentration (Product Category):** Largest cohort resides in 'Hardware' with 59.5 (18.1% of cohort total). Lead cohort exceeds 'Raw Metals' by 1.2x.
- **Unique Entities (SKU_Code) Distribution:** Distribution of Unique Entities (SKU_Code) of 400. Denominator rules and null exclusions applied deterministically.

DATASET INTEGRITY & STATISTICAL COVERAGE

Audited Records: 400 | Fully Populated: 400 (100.0%) | Duplicates: 0 | Missingness Exclusions Documented

VISUAL COHORT BREAKDOWNS & OPERATIONAL ACTION QUEUE

DISTRIBUTION BY PRODUCT_CATEGORY (Product_Category vs Mean Unit_Cost)

VOLUME BY WAREHOUSE_SITE (Warehouse_Site vs Volume)

METHODOLOGY, AUDIT TRAIL & METRIC DICTIONARY

METHODOLOGICAL POLICIES & AGGREGATION AUDIT TRAIL

Policy #1: All analytical aggregations are computed locally in-browser using full-precision raw IEEE 754 float values.

Policy #2: No intermediate rounded values are ever re-aggregated across groups, metrics, or export layers.

Policy #3: Non-additive metrics (ratings, GPA, percentages, unit prices) are aggregated via weighted or arithmetic mean/median only; summation is strictly forbidden.

Policy #4: Semi-additive metrics (stock on hand, account balances) are point-in-time state indicators and must not be aggregated over time dimensions.

Policy #5: Missing, null, or empty string values are excluded on a per-metric basis without corrupting adjacent columns.

Policy #6: Zero AI hallucination guarantee: 100% deterministic code compilation with verified three-way engine parity.

CANONICAL METRIC DICTIONARY & ADDITIVITY TIERS

METRIC LABEL	FORMULA SPECIFICATION	ADDITIVITY	EXCLUSIONS	CANONICAL VALUE
Total Records Volume	`COUNTROWS`	**ADDITIVE**	0 omitted	**400**
Unique Entities (SKU_Code)	`COUNTDISTINCT(SKU_Code)`	**ADDITIVE**	0 omitted	**400**
Mean Unit_Cost	`AVERAGE(Unit_Cost)`	**NON_ADDITIVE**	0 omitted	**54.73**
Median Unit_Cost	`MEDIAN(Unit_Cost)`	**NON_ADDITIVE**	0 omitted	**54.50**
Min Unit_Cost	`MIN(Unit_Cost)`	**NON_ADDITIVE**	0 omitted	**12.50**
Max Unit_Cost	`MAX(Unit_Cost)`	**NON_ADDITIVE**	0 omitted	**96.50**

INDEPENDENT MATHEMATICAL RECONCILIATION & REPRODUCIBILITY GUARANTEE

Source Hash: 4f480840 | Report Hash: b0949880 | Audit Timestamp: 2026-10-23T16:31:01.122Z

All metrics verified against independent Python float64 recalculation. Full raw precision preserved across all export targets.